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Multidimensional Poverty: A New Perspective on Measurement

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ABSTRACT *Amartya Sen's work on functionings and capabilities has transformed the understanding of poverty as a multidimensional issue, emphasizing capability deprivation. This paper distinguishes between poverty and capability deprivation, arguing that poverty results from insufficient access to resources, whether through markets or public provisions. We highlight that many existing multidimensional poverty indices, including those of the UNDP and Alkire and Foster, focus predominantly on capability deprivation and incorrectly assume that all individuals who experience it are poor. We argue that a comprehensive definition of poverty requires considering both capability and access to resources. Our proposed framework for measuring multidimensional poverty addresses these shortcomings and discusses its policy implications.*

KEYWORDS: Multidimensional poverty; capability deprivation; well-being

JEL CLASSIFICATION: I3; I32; D63; O3

1. Introduction

In the 2000 World Bank Development Report (WDR), poverty is defined as “the pronounced deprivation in well-being.” This definition prompts essential questions: How do we define well-being? What elements are necessary to ensure a decent level of well-being? Throughout the 1970s and 1980s, economic literature presented various approaches to understanding well-being, focusing on concepts such as basic needs, economic growth, quality of life, utility, welfare, and happiness. However, these approaches lack a cohesive framework for defining well-being.

Among the numerous conceptualizations, Amartya Sen's framework stands out as the most comprehensive. His innovative notions of “functionings” and “capabilities” articulate that functioning represents what individuals achieve in life, while capability refers to their freedom to achieve those outcomes. For Sen, the essence of well-being is deeply rooted in individuals' autonomy to determine the quality of their own lives. Thus, well-being is inherently multidimensional, contingent upon the capabilities individuals enjoy across various life domains.

The World Bank's definition of poverty aligns with Sen's insights, seeing it as a result of capability deprivation. A person is deemed poor when they lack fundamental capabilities. This

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definition enjoys considerable acceptance because it reflects the diverse deprivations individuals encounter.¹ However, a critical question emerges: Can poverty be addressed solely through the lens of capability deprivation?

In this paper, we propose a vital distinction between poverty and capability deprivation. We assert that poverty arises specifically when a lack of essential capabilities is linked to inadequate access to resources, whether through markets, public provisions, or other non-market avenues. Focusing solely on capability deprivation can obscure our understanding of who qualifies as poor. We contend that widely recognized measures of multidimensional poverty often fail to accurately capture the complexity of poverty because they primarily consider the deprivations people face, assuming that all individuals experiencing deprivation are poor. Instead, our analysis emphasizes that understanding the causes of deprivation—rooted in insufficient resources—is crucial for accurately measuring multidimensional poverty.

To further clarify our argument, we distinguish between “outputs” and “outcomes.” Outputs are directly influenced by individuals’ entitlements—derived from their resources or government assistance—while outcomes reflect ultimate achievements. Government policies aimed at reducing poverty influence outputs, which in turn affect outcomes; however, these interventions do not have a direct impact on the outcomes themselves. Therefore, we argue that poverty should be assessed through the lens of outputs, while the overall well-being of the population should be measured by outcomes.

In this paper, we introduce a novel methodology for measuring multidimensional poverty that accounts for the insufficiency of means available to individuals, essential for achieving their capabilities. By doing so, we aim to advance the academic discourse on poverty measurement, enhancing our understanding of its root causes and implications for policy intervention.

2. Relative and absolute poverty

In unidimensional poverty measurement, we identify individuals as poor based on a poverty line, which represents the income level necessary to meet basic needs. Once the poor are identified, we aggregate the extent of poverty they experience into an overall index. This aggregate index provides insights into both the percentage of the population that is poor and the intensity of their poverty, essentially asking how poor the poor are. Notable examples of aggregate poverty measures include Sen’s (1976) poverty measure and the Foster et al. (1984) class of poverty measures established in 1984.

Kakwani and Son (2016) propose two distinct approaches to measuring aggregate poverty. The first is the relative approach, which maintains that poverty remains unchanged if the incomes of all individuals and the poverty line increase or decrease by the same proportion. Most poverty measures in the literature are based on this relative framework. Conversely, the concept of absolute poverty measures is relatively new; it posits that poverty remains constant when the incomes of all individuals and the poverty line are adjusted by the same absolute amount. These absolute measures provide vital information on the monetary cost of eliminating poverty, making them more practical and easily comprehensible for policymakers.

In Section 6 of this paper, we elaborate on how we measure both relative and absolute poverty within a multidimensional framework. Both types of poverty measures play significant roles in shaping effective policies for eliminating poverty. However, the existing multidimensional poverty literature has yet to fully address this important issue.

Multidimensional poverty offers a comprehensive approach to measuring poverty by capturing the various deprivations individuals experience across multiple dimensions. However, this methodology introduces complex measurement challenges in practice. While progress has been made in identifying the poor and developing multifaceted poverty indices, significant challenges persist. Thorbecke (2008) highlights some of these unresolved issues, noting that multidimensional poverty measurement often relies on many arbitrary assumptions that can lead to inconsistent poverty assessments. In this paper, we critically review these unresolved issues and argue

that the focus on capturing capability deprivation tends to overlook the underlying causes of deprivation. We propose a new perspective on poverty measurement, suggesting an alternative analysis of multidimensional poverty that requires a lesser degree of arbitrariness.

3. Unresolved issues

In examining the complexities of multidimensional poverty, this paper identifies several unresolved issues that warrant further investigation. These issues are discussed as follows.

3.1. Essential capabilities

Measuring multidimensional poverty begins with identifying a set of essential capabilities that everyone in society should enjoy. The number of these socially desirable capabilities is referred to as the dimensions of poverty. A critical question arises: how many dimensions are necessary to adequately capture the multifaceted nature of poverty? Moreover, should we adopt a universal list of capabilities applicable to all countries, or should developing nations have their own distinct set of essential capabilities?

This topic has ignited a vigorous debate between Nussbaum (2011) and Sen (1993). Sen, the architect of the capability approach, refrains from endorsing a specific list of capabilities. He suggests that the selection of capabilities should emerge from democratic processes and social choice procedures, allowing the people to decide what is essential (Robeyns, 2005). In contrast, Nussbaum critiques Sen's position, arguing that to realize the full potential of the capability approach, he should advocate for a well-defined list of capabilities. Nussbaum has proposed a list of ten "central human capabilities," but these capabilities are so broad that they may not be practical for measuring multidimensional poverty in real-world situations.

As a result of this ongoing debate, no consensus has been reached on the specific list of essential capabilities. The life journey of individuals begins at birth and concludes at death, during which they experience numerous deprivations that can be difficult to quantify. Most academic papers analyzing multidimensional poverty typically focus on two or three attributes (Alkire & Foster, 2011; Bourguignon & Chakravarty, 2003). In comparison, the United Nations Development Programme's global multidimensional poverty index comprises ten indicators; however, it captures health deprivation through a solitary attribute, specifically the infant mortality rate. Consequently, the global MPI, which is employed by over 110 countries, fails to adequately represent the full spectrum of deprivations individuals face, rendering it a partial index of social deprivation.

Additionally, the challenge of selecting essential capabilities is further compounded by their evolution over time, driven by innovations such as digitalization and advancements in communication and transportation. For example, the necessity of cell phones for societal functioning has grown significantly. A person who cannot afford a cell phone may experience deprivation, as they struggle to engage in daily life without this essential tool.

In summary, determining the fundamental capabilities required to measure multidimensional poverty is a complex task. As society evolves and its needs change, our understanding of what constitutes the essential aspects of life for individuals in different contexts must also adapt.

3.2. Deprivation cut-offs

Suppose we have succeeded in selecting m dimensions, where $m \geq 2$, and there are n individuals within a population whose deprivation we aim to measure. From household surveys, we gather data on each person's achievement levels, denoted as x_{ij} . Here, x_{ij} reflects the achievement of the i th person in the j th dimension, indicating that a higher value of x_{ij} signifies greater well-being for the i th person in the j th dimension.

A person is considered deprived in the j th dimension if their achievement falls below a socially accepted minimum level of achievement in the j th dimension. Suppose society has determined that z_j is the minimum level of achievement every individual should attain for the j th dimension or capability, referred to as the deprivation cutoff. We identify the i th person as deprived of the j th capability if $x_{ij} < z_j$, and not deprived of $x_{ij} \geq z_j$.

To measure multidimensional poverty effectively, a deprivation cutoff must be determined for each dimension. However, the question arises: how can such cutoffs be meaningfully established across diverse dimensions? In the unidimensional approach, setting a poverty line—particularly within the income domain—presents significant challenges. While Kakwani (2011) introduced an innovative model for defining poverty thresholds related to food and non-food needs based on consumer theory, establishing thresholds across varied dimensions of well-being remains complicated.

Currently, no standardized methods exist for determining deprivation cutoffs for essential capabilities needed to identify poverty across different dimensions. For instance, how does one ascertain an appropriate threshold for life expectancy at birth or infant survival rates? Is it reasonable to define an acceptable norm as the survival of 700 out of 1,000 infants born? Arriving at a societal consensus on such standards proves difficult, especially when approached democratically. Similarly, is four years of education adequate for functioning effectively in society? Unfortunately, there is no definitive answer to this question. The reality is that most research on multidimensional poverty has relied on establishing thresholds on an ad hoc basis, lacking a systematic approach to determining these critical norms.

3.3. Dashboard vs single indices of poverty

In 2011, The Journal of Economic Inequality published a special issue featuring a forum dedicated to exploring the best methods for measuring the various dimensions of deprivation that individuals experience. Four prominent scholars—Ravallion, Ferreira, Alkire, and Foster—contributed insightful papers addressing this complex issue. Ravallion proposed a ‘dashboard approach,’ which involves measuring deprivation separately for each dimension of poverty rather than aggregating these deprivations into a single scalar metric. This approach allows for a clearer assessment of performance across different dimensions of deprivation. For instance, if the health dimension reveals poor outcomes, targeted interventions can be directed towards the health sector to enhance performance and improve health results.

Despite its strengths, the dashboard approach has two significant limitations. First, it does not provide insights into the overall incidence of poverty or the effectiveness of a country’s efforts in reducing poverty comprehensively. Second, because the dashboard is based on marginal distributions, it overlooks the joint distribution of various dimensions, which is crucial for understanding the interconnectedness of deprivations.

Empirical research by Duclos et al. (2006) highlights the importance of considering the joint distribution of deprivations when evaluating poverty. They argue that the correlation between deprivations across different dimensions is significant; the strength of this correlation can notably affect poverty comparisons. As a result, the literature has increasingly favored single indices of multidimensional poverty.

Nonetheless, a disagreement persists between Ravallion and Ferreira regarding the use of a single poverty index. Ravallion contends that a single index, which relies on numerous ad-hoc assumptions, cannot adequately capture the complexity of multidimensional poverty. In contrast, Ferreira argues that a multivariate single poverty index is preferable since it harnesses the additional information contained in the joint distribution of deprivations. Ultimately, the forum did not resolve this ongoing debate, leaving room for further exploration and discussion on this critical issue.

3.4. Trade-offs between dimensions

When we aggregate deprivations across different dimensions into a single poverty index, it is essential to consider the potential interactions between these dimensions. If the dimensions are substitutes, a person may be able to trade off a deprivation in one area for a benefit in another. For instance, could one accept a trade of sacrificing one year of education in exchange for two additional months of life? Such tradeoffs may seem unusual.

However, these trade-offs are unavoidable when estimating aggregate poverty indices. The single index inherently assumes a certain level of trade-off among dimensions. In practical contexts, we often lack the means to accurately estimate these trade-offs using real-world data, leading us to rely on assumptions. For example, in their seminal work, Bourguignon and Chakravarty (2003) introduce a parameter, θ , in the poverty function to quantify the trade-offs between income and education. Specifically, when θ equals 1, it indicates perfect substitutability; conversely, higher values of θ suggest reduced substitutability. They provide multidimensional poverty estimates based on various values of θ , highlighting our uncertainty regarding the most appropriate value. This uncertainty has significant implications for deriving accurate poverty measures and, consequently, for shaping effective policy responses.

3.5. Deprivation weights

Consider two individuals, A and B. Person A is illiterate and struggles to function effectively in society, while person B has been diagnosed with cancer. Each experiences distinct forms of deprivation, leading to different types of suffering. This raises an important question: can we compare the deprivation experienced by A and B? Is A suffering more than B, or vice versa? If so, to what extent can we quantify this difference? A straightforward response might simply be that we do not know. Estimating a social deprivation index necessitates quantifying deprivation across various dimensions.

To illustrate this, let us consider a vector, $w = (w_1, w_2, \dots, w_m)$, representing deprivation scores. These scores reflect the relative extent of deprivation experienced by individuals in different areas of their lives. Importantly, the sum of the weights assigned to these dimensions must equal 1. These weights indicate the importance attributed to each dimension; a higher weight signifies a more significant degree of deprivation.

The methodology proposed by Alkire and Foster (2011) is crucial for identifying poverty and measuring the social deprivation index. However, the process of determining these weights is inherently subjective, as there is no objective method for establishing them. Researchers frequently adopt weights on an ad hoc basis. While Alkire and Foster argue that this approach provides flexibility and can be adjusted for specific purposes, we contend that relying on ad hoc weights can lead to inconsistent and unreliable poverty indices.

Pattanaik and Xu (2018) bring an important perspective to this issue, stating that “Even in a given society, there may be distinct culturally identifiable groups that have different sets of weights for various dimensions.” This observation is particularly relevant in diverse societies, such as India, where cultural groups may have varying perspectives on the importance of different deprivation dimensions. This prompts us to consider which set of weights should apply to society as a whole, a question that Pattanaik and Xu do not resolve.

The challenge of comparing poverty status across different social groups becomes pronounced when there are different deprivation weight structures. For instance, consider two individuals suffering from the same disease, such as cancer, but from different cultural groups. If these individuals are assigned different weights based on their social context, it becomes methodologically problematic to compare their levels of social deprivation.

Similarly, comparisons between rural and urban populations within a country highlight these complexities. Typically, rural populations tend to experience higher levels of poverty than their urban counterparts, and the disparity in living conditions can be stark. As discussions about poverty emerge in policy debates, a crucial question arises: can we meaningfully evaluate poverty between these two groups from a multidimensional perspective?

Unique needs exist for both populations, which means that the importance placed on various deprivation dimensions will differ, leading to variations in the assigned weights. This variability complicates the development of social deprivation indices that accurately reflect the multidimensional nature of poverty in both rural and urban areas. Consequently, this limitation hinders our ability to determine whether the poverty gap is narrowing or widening, highlighting a significant flaw in the methodologies currently used to assess multidimensional poverty.

Further compounding this issue are the dual deprivation cut-offs proposed by Alkire and Foster (2007, 2011). These cut-offs aim to identify multiple forms of deprivation simultaneously, but they also rely on the subjective weighting of dimensions. This dependency can exacerbate the challenges of creating reliable poverty indices, particularly in culturally diverse societies where individual experiences of deprivation can vary widely.

3.6. *Dual deprivation cut-offs*

Alkire and Foster (2007, 2011) have made a significant contribution to measuring multidimensional poverty. Their methodology has received widespread recognition and praise. The poverty indices they developed utilize dual cutoffs, which are critically important. Given the popularity of the dual cut-offs, reviewing them is timely and appropriate.

Consider x_{ij} as the achievement of the i th person in the j th dimension, where j ranges from 1 to m , with m representing the total number of dimensions. Let z_j denote the minimum level of achievement that every individual must attain in the j th dimension or capability, which serves as the first deprivation cutoff. Additionally, let p_{ij} represent the probability of the i th person being deprived in the j th dimension, defined by the following criteria:

$$p_{ij} = 1, \text{ if } x_{ij} < z_j \\ = 0, \text{ otherwise.}$$

Let the deprivation score of the i th person in the j th dimension be represented as follows:

$$w_{ij} = p_{ij}w_j$$

where w_j is the deprivation score for the j th indicator, which is constant for all individuals. The total deprivation score of the i th person is given by:

$$w_i = \sum_{j=1}^m p_{ij}w_j.$$

Note that the second cutoff is referred to as the poverty cutoff. The probability of the i th person being identified as poor is defined by the following equation:

$$q_i = 1 \text{ if } w_i > \frac{1}{3} \\ = 0, \text{ otherwise}$$

which gives the number of multidimensional poor as

$$q = \sum_{i=1}^n q_i.$$

Only the poor experience deprivation, while the non-poor do not. Let n represent the number of people in the population; the per capita deprivation score of the population is defined by

$$M_0 = \frac{1}{n} \sum_{i=1}^q w_i = \frac{1}{n} \sum_{i=1}^q p_{ij} w_j$$

which is the Alkire-Foster proposed multidimensional poverty index.

The headcount ratio (H) represents the proportion of individuals in the population who are considered multidimensionally poor: $H = \frac{q}{n}$. The average level of deprivation experienced by the poor is represented by: $A = \frac{\sum_{i=1}^q w_i}{q}$.

It can be easily seen that $M_0 = HA$ is similar to the poverty gap ratio in unidimensional poverty. M_0 can be calculated for both cardinal and ordinal measures of individuals' achievements. If the variables are all cardinal, the Alkire and Foster methodology allows us to derive the generalized FGT measures of poverty, which provide insights into the depth and severity of poverty in a multidimensional context.

A general class of multidimensional FGT measures can be denoted by M_α , where α indicates aversion to poverty. When $\alpha = 1$, we measure the multidimensional poverty gap ratio. Similarly, when $\alpha = 2$, we measure the severity of multidimensional poverty. However, appealing features have been challenged by Pattanaik and Xu (2018), who argue that when individual dimensional deprivations are measurable in cardinal terms, identifying the poor may not always be feasible.

3.7. Ad-hoc assumptions

The literature on multidimensional poverty has overlooked critical issues arising from the arbitrary assumptions made in constructing multidimensional indices. For example, the M class of poverty measures assumes a poverty cut-off point of over one-third, which is an entirely arbitrary figure. By merely increasing this cut-off, one can reduce the reported incidence of poverty. Surprisingly, many countries have adopted the M class of poverty indices without questioning these arbitrary assumptions.

In this section, we have highlighted numerous unresolved issues related to multidimensional measurement. The analysis suggests that aggregating dimensions of deprivation while relying on arbitrary assumptions can lead to inconsistent measures of poverty. Nonetheless, the popularity of these indices continues to grow; the UNDP has adopted and reported on them for more than 110 countries.

What should we do about this situation? In the rest of this paper, we will attempt to address these concerns.

4. Poverty and capability deprivation

The basic needs approach is widely used in measuring poverty, a concept first articulated by Rowntree (1901). This approach identifies individuals as poor if they experience deprivation due to their inability to access essential goods and services necessary for living. While we refer to this as the basic needs approach, deprivation can extend beyond these essentials, affecting areas like health and education. It is crucial to distinguish between deprivation stemming from inadequate income for basic needs and other forms of deprivation, as this distinction is fundamental to understanding poverty.

Generally, the higher a person's income, the greater their access to commodities and services, which ideally enhances their quality of life. However, access to resources is merely a means to an end. Individuals can still experience significant deprivations in various life aspects despite possessing resources. Therefore, defining poverty solely in terms of available means is inadequate. According to Sen (1993), a comprehensive understanding of poverty must include capability failures that lead to deprivations across multiple dimensions of well-being. This

perspective has gained widespread acceptance in multidimensional poverty literature, resulting in the development of indices based on the capability failures encountered by individuals.

One may question whether poverty can be described purely as capability deprivation. Consider the case of the wealthiest person in the country, who suffers from an incurable disease that renders them non-functional and causes immense pain. While this represents severe capability deprivation, it challenges the conventional notion of poverty—labeling such an individual as “poor” seems counterintuitive.

To illustrate, let us examine two individuals experiencing malnutrition. Person A, driven by religious beliefs, regularly fasts and suffers from malnutrition. In contrast, Person B cannot afford sufficient food. While both individuals face capability deprivation, we would not classify A as poor due to her voluntary choices, whereas B’s situation reflects poverty caused by a lack of resources. Thus, poverty arises specifically from capability deprivation linked to insufficient means.

Moreover, many income-rich individuals may suffer from health issues like obesity due to poor lifestyle choices. Despite enduring serious health problems, these individuals are not classified as poor, as their deprivation stems from choices rather than lack of resources.

A relevant example is King Charles of Great Britain, who was diagnosed with cancer and subjected to treatment. Despite his suffering, we do not consider him poor due to his position and wealth. Conversely, an ordinary person diagnosed with cancer who cannot afford treatment would undoubtedly be considered poor, as their suffering is a direct result of insufficient resources.

In measuring poverty, it is essential to differentiate between poverty and general capability deprivation. Capability deprivation may arise from various factors, but poverty specifically concerns the inadequacy of resources required to meet socially determined basic capabilities. While some individuals may experience capability deprivation, this does not automatically classify them as poor.

Defining poverty from the capability perspective cannot exclude considerations of income or available resources. The capability to function is influenced by various factors, but when dysfunction results from inadequate resources, the individuals affected are undeniably poor.

Consider a hypothetical scenario in which a person suffers a stroke and requires urgent medical attention to survive. If they lack the financial means to access healthcare, their death would be a capability failure attributable to their insufficient resources, representing a clear case of acute poverty. Such situations are prevalent in the real world. In contrast, if a wealthy individual suffers a stroke but has all the necessary resources, society views this as unfortunate luck rather than poverty.

In conclusion, understanding poverty requires clear distinctions between capability deprivation and poverty itself, with a focus on the resources needed to sustain basic capabilities.

5. Outputs and outcomes

Multidimensional poverty relates to the failure of basic capabilities, which can be assessed through two types of indicators: outputs and outcomes. Outputs, such as access to health care services—including prevention, diagnosis, treatment, and management of illnesses—are distinct from outcomes, like life expectancy. Similarly, when we consider education, outputs include access to schooling for all school-age children, regardless of their families’ financial situations. In contrast, outcomes pertain to what students learn, such as literacy and attainment of educational milestones, which can be influenced by factors like parental education and occupation.

Economic resources play a crucial role in shaping these outputs. Individuals’ income and resources provided by government social programs directly affect their access to essential services. However, simply generating outputs is not enough; they must be designed to maximize positive outcomes.

This leads to a key question: Should multidimensional poverty be measured by outputs or outcomes? While outcomes may intuitively seem more relevant as they reflect societal progress in well-being, they often stem from a variety of influences, making them harder to attribute to specific causes. Conversely, multidimensional poverty stems from inadequate access to resources that enable individuals to enjoy essential capabilities.

The World Bank defines poverty as the pronounced deprivation of well-being, but this definition overlooks the underlying causes of such deprivation. For instance, if a person cannot access healthcare due to financial constraints, this represents output deprivation linked to poverty. In contrast, consider a millionaire with knee pain due to arthritis. His situation is a result of individual circumstances, representing outcome deprivation. If he could afford a knee replacement, that would be an outcome; if not, it highlights the limits of his economic means.

Thus, distinguishing between output and outcome deprivation is essential. Output deprivation is primarily driven by poverty, while outcome deprivation results from a broader range of unpredictable factors. Therefore, measuring poverty should focus on outputs.

Outputs should ensure food security, adequate clothing, shelter, and access to essential amenities, such as electricity and clean water. Additionally, individuals should have access to basic healthcare and quality education, enabling them to become literate adults.

The relationship between outputs and outcomes is significant, as outputs directly influence metrics like mortality rates, literacy rates, and life expectancy. Government policies cannot directly control outcomes because they are caused by various individual factors. However, governments can implement interventions that are outputs designed to achieve desirable outcomes. Impact evaluation studies are commonly conducted to measure their impact on outcomes.

Examining the healthcare systems of the USA and Australia illustrates this point. Despite the USA spending nearly twice as much per capita on healthcare, it has a life expectancy about five years lower than Australia. This disparity reveals that Australia achieves better health outcomes at a lower cost, underscoring the importance of efficiency and equity in generating outputs, which is vital for alleviating multidimensional poverty.

6. How should We measure multidimensional poverty?

Alkire's contributions to the definition and measurement of multidimensional poverty have been remarkable.² She published a rejoinder: *Poverty and Well-being* in 2006, in which she argued, "The problem with defining poverty according to its cause is not only that it is difficult to establish causality, but it also may not be necessary: basic capabilities can be expanded by public action regardless of causality." Alkire's assertion encompasses three key points: (i) the challenge of establishing causality, (ii) the claim that causality is unnecessary, and (iii) the belief that public action can enhance basic capabilities independent of causative factors. We have reservations about all three points.

To illustrate, consider a hypothetical individual diagnosed with cancer. This person suffers tremendously if they lack access to available medical services. Indeed, access to healthcare is crucial for reducing capability deprivation. To obtain such services, one typically needs either personal financial resources or government-provided subsidies. Without the means to access medical care, this individual may experience severe deprivation. This scenario highlights the significance of causality: access to medical services directly impacts an individual's capability. Thus, we cannot dismiss causality as unnecessary. Furthermore, the very public action that benefits a cancer patient—namely, providing medical services—relies on the financial means that these individuals must possess.

The first step in measuring multidimensional poverty is to clearly define the essential capabilities that everyone should have the right to enjoy. We classify a person as poor if they are unable to secure the following four capabilities:

1. Food security
2. Health security

3. Education security
4. Security of basic living

These broad categories of basic capabilities are closely linked to the concept of poverty. While other capabilities, such as protection from crime, access to the judiciary, employment opportunities, and protection from natural disasters, are important, they are relatively less critical for the direct measurement of poverty.

The four basic capabilities listed above serve as outputs and can be quantified within the framework of income. The per capita average costs associated with each capability can be estimated using household expenditure surveys, government statistics, or market prices for goods and services. We refer to these costs as poverty cutoffs, which do not necessitate arbitrary dual thresholds for individual capability deprivation and can be derived from various data sources. Governments can alleviate some of these costs through social welfare programs, which lessen the burden on individuals. Thus, public actions can significantly enhance people's security related to these four capabilities, effectively reducing poverty.

Let us denote the poverty cutoffs associated with food, health, education, and basic living security as z_f, z_h, z_e , and z_l . A household is identified as multidimensionally poor if its per capita income from various sources, denoted as x , is less than the total of these four cutoffs, expressed as $z = z_f + z_h + z_e + z_l$. The probability distribution function $F(x)$ can be used to determine the proportion of individuals who are multidimensionally poor due to insufficient income to achieve these basic capabilities, $H = F(z)$. This measure is known as the headcount ratio, although popular, it does not capture the intensity of poverty experienced by individuals.

To address intensity, there are two approaches as discussed in [Section 2](#): relative and absolute. The relative approach maintains that the intensity of poverty remains unchanged if the income of the poor and the poverty line rise or fall by the same percentage. In contrast, the absolute approach asserts that intensity remains constant with absolute changes in income and the poverty line.

Sen (1976) developed his relative poverty measure using an axiomatic approach, which became a new mantra for poverty measurement. His measure is given by

$$S_R = \frac{z - \mu_z(1 - G_Z)}{z} \quad (1)$$

where μ_z represents the mean income of the poor, and G_Z is the Gini index of the poor. $\mu_z(1 - G_Z)$ measures the social welfare of the poor, which is a homogeneous function of degree one. That implies that if the incomes of the poor increase or decrease by the same proportion, their social welfare also increases by the same proportion. It then immediately follows that S_R is a relative measure of poverty. This measure is interpreted as the welfare gap of the poor from the poverty line as a proportion of the poverty line.

Sen did not propose an absolute poverty measure, but it is easy to derive an absolute poverty measure given by $S_A = zS_R$, satisfying the requirement that the measure remains invariant when the incomes of the poor and the poverty line change by the same absolute amount. Its attractive feature is that it informs how much monetary cost is needed to eliminate poverty.

We have identified four key areas of capability deprivation that warrant attention. Policymakers would benefit from measuring poverty within each of these categories: food, health, education, and living conditions. The methods previously discussed can be readily adapted to assess poverty in these four domains. The headcount ratio for the j th category can be denoted as $H_j = F(z_j)$, where j takes the values f, h, e , and l . The headcount ratio for the multidimensional poverty can be expressed as $H = f(H_f, H_h, H_e, \text{ and } H_l)$, providing a holistic view of poverty across capabilities. Similarly, Sen's aggregate multidimensional poverty can be formulated by $S(z) = f(S(z_f), S(z_h), S(z_e), \text{ and } S(z_l))$, enabling us to analyze the overall

impact of poverty on different capabilities. These aggregate functions are crucial for calculating how poverty affects each capability area. To utilize them effectively, we will need to establish specific poverty cut-offs for each capability, which we will explore using real data in the following section.

7. Poverty cut-offs for insecurities

7.1. Food security

Food is a fundamental necessity for all human beings. Extreme food insecurity occurs when people are uncertain about when or if they will have their next meal and lack the means to purchase the food they desire. As a result, they experience hunger, which we consider an extreme form of poverty. In contrast, a more common form of poverty arises when individuals cannot afford food that meets their nutritional needs, leading to malnutrition. This type of food insecurity is reminiscent of the discussions held at the 2009 World Summit on food insecurity.

Maintaining good health involves consuming a balanced diet that provides an adequate intake of calories, proteins, fats, and carbohydrates. The food poverty line represents the minimum per capita cost required for a family to afford a balanced diet for all its members. A family is considered food secure if its per capita income consistently meets or exceeds this cost, $y \geq z_f$.

It is crucial to define what constitutes a balanced diet for families of various sizes and to estimate its associated costs. According to Kakwani and Son (2022), a balanced diet for a family should provide approximately 2,100 kilocalories per person per day. Ideally, this energy intake should derive from 60–70% carbohydrates, 15–30% fats, and 10–15% protein.

Typically, poorer families tend to rely on food baskets that consist primarily of carbohydrates due to their lower costs. However, this can lead to malnutrition, emphasizing the need for all families to have sufficient resources to ensure that every member consistently has access to a balanced diet.

Kakwani and Son (2022) also offer a methodology for estimating the per capita cost of a balanced diet, although reproducing their methodology is not necessary for this discussion. Measuring food security is a vital development goal, and we have incorporated it as a key element within the broader context of multidimensional poverty.

The United Nations Development Programme (UNDP) utilizes the global multidimensional poverty index (MPI), which includes the body mass index (BMI) as a measure of undernutrition. However, it is important to note that a low BMI is not solely indicative of a lack of financial means. Factors such as anorexia—an eating disorder that can affect individuals from wealthier backgrounds—also contribute. Additionally, BMI fluctuates due to differences in individual metabolism. Thus, low BMI may reflect an outcome rather than serve as a suitable measure of poverty.

7.2. Health security

Inequity in health delivery services is a pressing issue globally, particularly in developing countries. Millions of individuals suffer from preventable illnesses, including both infectious and chronic diseases, leading to unnecessary deaths. This suffering often stems from a lack of access to adequate medical services.

We define a person as health secure if they have continuous access to all basic health services, and health insurance serves as a vital means of accessing these services. In some countries, universal healthcare exists, where governments provide essential healthcare to all citizens based on their health needs. However, the absence of universal coverage leaves many populations without the medical attention they require. Numerous studies indicate that government health services tend to favor the well-off, while the poor, who typically have greater health needs, are left

underserved. Additionally, severe budget constraints in nearly all developing countries prevent many low-income individuals from obtaining even the minimum quality of health services. User fees for government-provided services further exacerbate the situation, as many impoverished individuals cannot afford these costs, leading them to forgo necessary care and ultimately suffer from avoidable illnesses or even death. Thus, health poverty can become a persistent challenge in developing nations.

The health costs that families face can be summarized by three key factors: insurance costs, out-of-pocket expenses, and the cost of medications. When a family purchases health insurance, they typically gain access to essential medical services, including consultations with general practitioners and specialists, medical procedures, surgeries, diagnostic tests, and X-rays. However, in reality, health insurance often does not cover all expenses—particularly those related to dental care—necessitating additional out-of-pocket costs. The costs of medications can also be substantial, placing a further burden on families.

To determine whether a family is health secure, we can define a threshold: if their total per capita income consistently exceeds the per capita health costs they face, they are classified as health secure. Conversely, families that fall below this threshold are deemed poor in health. This practical approach to defining health poverty differs from the UNDP's global MPI, which categorizes a family as health-poor if any child has died within the last five years. Relying on this singular indicator fails to encompass the multitude of health-related deprivations that individuals face.

People endure a variety of chronic illnesses, such as diabetes, cancer, heart disease, strokes, depression, and dementia. Moreover, the elderly experience significant suffering in their later years. Many become frail and unable to care for themselves, often requiring intensive medical assistance. Without the financial means to access the necessary care, they find themselves in severe poverty. These are the real issues of poverty that the existing multidimensional poverty literature overlooks. As a result, the global MPI appears disconnected from the true poverty issues that exist in the world today.

7.3. Education security

Equitable access to education is closely linked to poverty reduction. Education plays a crucial role in lifting people out of poverty by equipping them with the skills needed to enhance their employability and productivity. All school-age children are required to attend school, regardless of their family's economic situation. We define a family as enjoying educational security if they have the financial means to always educate all their school-age children. In many developing countries, some children do not attend school because they are engaged in labor to support their families economically. This results from poverty, leading to significant deprivation for the children involved.

Families bear substantial costs to support their children's education, which vary depending on the level of education—primary, secondary, or tertiary. These costs can include school fees, private tuition, books, notebooks, pencils, pens, computers, school uniforms, iPads, school lunches, and other school supplies. Consider a family's per capita income is represented as y , and the total per capita cost of educating all children is denoted as z_e . We identify a family as education secure if $y_i \geq z_e$ at all times. Conversely, a family is considered education-poor if it lacks educational security.

All school-age children possess a fundamental right to education. However, when families face financial barriers that prevent them from fulfilling this right, it becomes the government's responsibility to provide support. In Brazil, the Bolsa Família Program (BFP) stands out as a leading initiative in social protection, recognized globally as the most effective conditional cash transfer program (CCT).

The Bolsa Família operates by offering cash transfers to low-income families under specific conditions. In exchange for financial assistance, these families must ensure their children attend school regularly and undergo routine health checkups. This program serves two vital purposes: it offers immediate economic relief to impoverished households while also addressing long-term poverty reduction by incentivizing education and health care for children.

Extensive evaluations of social programs, such as the Bolsa Família, have employed rigorous impact evaluation techniques. The findings consistently demonstrate that CCT programs significantly reduce both monetary deprivation and educational poverty. Furthermore, these initiatives lead to improved educational outcomes, including higher literacy rates and increased educational attainment among beneficiaries. Through such programs, Brazil is making strides toward a more equitable and educated society.

7.4. Security of basic living

Poverty is fundamentally linked to how people live and whether their basic needs are met. Historically, concerns centered around three essential needs: food, shelter, and clothing. However, in modern times, the understanding of basic needs has significantly evolved, now encompassing a broader range of goods and services necessary for individuals to function effectively in society. Simply having food, shelter, and clothing is no longer sufficient.

The global MPI identifies six key indicators of deprivation in basic living conditions:

1. Electricity: The household has no access to electricity.
2. Sanitation: The household's sanitation facilities are inadequate.
3. Water: The household lacks access to safe drinking water.
4. Floor: The household is equipped with dirt, sand, or dung floors.
5. Cooking Fuel: The household cooks use dung, wood, or charcoal.
6. Asset Ownership: The household does not possess essential assets such as a radio, TV, telephone, bicycle, motorbike, refrigerator, or car/truck.

These indicators predominantly pertain to rural and remote areas. In urban metropolitan regions, it is uncommon for households to lack electricity or have inadequate sanitation with floors made of dirt, sand, or dung. Furthermore, very few urban households rely on dung or wood for cooking. Those enduring the deprivations listed would be considered impoverished when compared to urban residents. However, a crucial question arises: Can individuals living under such challenging conditions still function effectively within their society? The answer is likely affirmative; in rural areas, many residents share similar living conditions, mitigating feelings of deprivation. Thus, when evaluating overall deprivation in a country, these six indicators may not accurately reflect the true state of poverty.

In constructing the global MPI, each of these six indicators is assigned an equal weight of 5.6%. This means that if a household classified as poor owns a radio or car, its perceived poverty is reduced by 5.6%. However, the impact of ownership—whether of an unused old radio or a functioning car—is treated equivalently within this framework. This approach raises concerns, as intuitively, the importance of various indicators should not carry equal weight in the eyes of individuals affected by poverty.

Moreover, many critical expenditures that underpin basic living are overlooked by the global MPI. Important categories missing from the index include transportation and communication, clothing and footwear, laundry and cleaning necessities, energy consumption, personal care items, rental costs for adequate housing, and baby care expenses. Household expenditure surveys provide insights into these costs while also detailing the age and sex of each household member, enabling a more accurate estimation of individual basic needs. By aggregating the needs of all household members, we can calculate the per capita expenditure, which would then serve as the poverty line for basic living.

A family is considered to enjoy security in basic living if its expenditures consistently meet or exceed this poverty line. Conversely, a household that fails to achieve this level of financial security would be categorized as poor in terms of basic living standards.

In conclusion, while the global MPI sheds light on poverty through specific indicators, it is imperative to recognize and address the multifaceted nature of what constitutes a basic living standard in today's world. A comprehensive understanding of poverty requires taking into account a broader spectrum of needs and expenditures to accurately reflect individuals' living conditions and experiences.

8. Concluding remarks

The discourse surrounding multidimensional poverty has primarily centered on measuring the capability deprivations experienced by individuals across various dimensions. Specifically, the emphasis has been on developing comprehensive indices that capture the extent of overall capability deprivation within a population and track changes in this deprivation over time. This body of literature has made considerable strides in transcending traditional unidimensional poverty measures, allowing for a more nuanced understanding of the complex nature of deprivation.

Multidimensional poverty is fundamentally viewed as a failure to achieve essential capabilities, marking a significant shift from the income-centric approach that merely suggests increasing incomes for those below the poverty line. The capability approach focuses on well-being rather than economic means, prioritizing the myriad deprivations individuals face in different aspects of life.

For instance, if an analysis reveals that a society is experiencing serious health-related deprivation, we must ask ourselves: how can we address this? A logical response is to evaluate the efficiency and equity of health service delivery. Efficiency encompasses the presence of qualified healthcare providers and the necessary medical equipment to effectively treat chronic illnesses, while equity addresses the accessibility of these services—highlighting that prohibitive costs can prevent the poor from seeking necessary care, thus exacerbating capability failures.

Understanding the root causes of capability failure is crucial before implementing policies aimed at alleviating it. The literature on multidimensional poverty seeks to identify the broader experiences of deprivation across the population. It is important to note that deprivation can afflict both the affluent and the impoverished; however, poverty, in this context, specifically refers to deprivation stemming from a lack of means. As an illustration, while a wealthy individual may face a serious health crisis, labeling him as “poor” due to his condition would be fundamentally misleading.

To effectively reduce or potentially eliminate multidimensional poverty, we must thoroughly assess the efficiency and equity of the services available within a country. Enhancing service equity should be a priority, ensuring that marginalized populations have the means to access health care, receive basic education, and meet their fundamental needs, thereby preventing situations of hunger and vulnerability. Many nations have implemented safety net programs aimed at providing social support to the impoverished, helping to shield them from acute deprivation.

Ultimately, addressing multidimensional poverty requires not only a robust understanding of its complexities but also a decisive commitment to empowering the vulnerable through equitable access to essential services.

Notes

1. See Atkinson (2003), Bourguignon and Chakravarty (2003), Tsui (2002), Dutta, Pattanaik, and Xu (2003), Alkire and Foster (2007, 2011), Alkire and Santos (2010, 2014), Chakravarty (2006), Deutsch and Silber (2005), Duclos, Sahn, and Younger (2006), and Maasoumi and Lugo (2008), among others.

2. Her book, published in 2015 with Foster, Seth, Santos, Roche, and Ballon, titled *Multidimensional Poverty and Measurement*, is a significant contribution to the understanding of multidimensional poverty.

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